**Math 242, S20** **Name:** Answer

**Quiz 3**

**Instructions:** **There is totally 1 problem. You must show all of your work to receive a credit for a correct answer.**

**Problem 1.** (10 points) Find the particular solution of the initial value problem:

$$x\frac{dy}{dx} = 2y + x^3 \sin x \cos x, \qquad y(\pi) = 2.$$

(*First check if the differential equation is given in the standard form of linear first-order differential equation*)

First transform the equation to the standard form

$$\frac{dy}{dx} - \frac{2}{x}y = x^2 \sin x \cos x$$

We have $P(x) = -\frac{2}{x}$, $Q(x) = x^2 \sin x \cos x$

Calculate the integrating factor

$$\rho(x) = e^{\int P(x)dx} = e^{\int -\frac{2}{x}dx}$$

$$= e^{-2\ln|x|} = \frac{1}{x^2}$$

Multiply $\rho(x)$ to both sides of the standard equation

$$\frac{1}{x^2}\frac{dy}{dx} - \frac{2}{x^3}y = \left(\frac{1}{x^2}\right)x^2 \sin x \cos x$$

$$\frac{d}{dx}\left[\frac{1}{x^2}y\right] = \sin x \cos x$$

$$\frac{1}{x^2}y = \int \sin x \cos x\, dx$$

$$\frac{y}{x^2} = \frac{1}{2}\sin^2 x + C$$

$$\Rightarrow y(x) = x^2\left(\frac{1}{2}\sin^2 x + C\right)$$

$y(\pi) = 2 \Rightarrow$

$$\pi^2\left(\frac{1}{2}\sin^2 \pi + C\right) = 2$$

$$\Rightarrow \pi^2 C = 2$$

$$C = \frac{2}{\pi^2}$$

Thus we obtain

$$y(x) = x^2\left(\frac{1}{2}\sin^2 x + \frac{2}{\pi^2}\right)$$